

The State Conditional Cross Sectional Multilayer Perceptron

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Abstract

This document describes the state conditional cross sectional multilayer perceptron for monthly portfolio construction. The model encodes each security from its characteristics, couples the securities through a pooled cross sectional context, and is fitted under a maximum Sharpe ratio regression objective, and it conditions the characteristic to score map on the prevailing market state through feature wise modulation. We develop the full model, motivate the conditioning from the time variation of factor premia, present the modulation formally, explain why an interaction with security level features is the only channel through which a state signal can survive the portfolio construction, and describe the construction of the state variables.

1 Overview and motivation

Factor premia are not constant through time, and a portion of their variation is predictable from observable conditioning variables. A model that maps characteristics to scores through a single function that is identical in every month cannot exploit this variation, because it applies the same ranking rule in every state of the world. The present model is therefore designed to let the state of the market change which characteristics matter, so that the ranking rule adapts to the conditions under which it is applied.

2 The cross sectional perceptron and the data pipeline

The conditioning is placed on a core network and data pipeline that we now set out. The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period, which makes the selection causal and algorithmic. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are set to the median before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped. Each security is encoded to an embedding z_i , a pooled context is formed as the average embedding across the month, and the score is read from a head on the concatenation of the embedding and the context. The network is fitted under the maximum Sharpe ratio regression loss $(1 - \sum_i w_i r_i)^2$ on a rolling window of at most sixty months, with a fifty epoch cold start, ten epoch warm starts thereafter, three seeds, and seed averaging of the weights normalised to unit gross exposure.

3 Feature wise modulation by the market state

Let $s \in \mathbb{R}^p$ denote the vector of standardised state variables for a given month. A single linear map turns the state into a scale and a shift for each dimension of the per stock embedding,

$$[\gamma, \beta] = W_{\text{film}} s, \tag{1}$$

and the embedding is modulated dimension by dimension,

$$z_i \leftarrow z_i \odot (1 + \gamma) + \beta, \quad (2)$$

where $\odot$ is the elementwise product and $\gamma, \beta \in \mathbb{R}^d$ are shared across the securities of the month. This is feature wise linear modulation. The state does not enter the score additively at the security level, but instead rescales and shifts the coordinates of the embedding, so it changes the relative importance of the learned features rather than the overall aggressiveness of the book. The modulation map is initialised at zero, so that $\gamma = \beta = 0$ at the start of training and the model begins exactly as the unconditioned perceptron. The conditioning is learned only if it earns its place, and it adds on the order of one thousand parameters to a network with tens of thousands.

4 Why the interaction is necessary

The modulation is designed to act on the embedding coordinates rather than as a level effect, and the reason is essential to the construction. The traded weights are demeaned across securities and normalised to unit gross exposure. Any signal that is additive or multiplicative at the level of the whole month is therefore annihilated by the construction itself, because adding a constant to every score leaves the demeaned and normalised weights unchanged, and scaling every score by a common factor leaves the normalised weights unchanged as well. It follows that a state signal can affect the traded book only if it interacts with security level features so as to change the relative ranking of the securities. Feature wise modulation provides exactly such an interaction, since a state driven rescaling of an embedding coordinate alters the ranking whenever securities differ on that coordinate. We can conclude that the dimension wise form of the modulation is not incidental but is the only form through which the conditioning can survive.

5 Construction of the state variables

The conditioning variables are all known at the end of the formation month. The state comprises the realised volatility of the market over twenty one and over two hundred and fifty two trading days, the trailing twelve month market return, the cross sectional dispersion of trailing one month returns, the value spread measured as the ninetieth minus the tenth percentile of book to market, and the trailing twelve month return of a momentum factor constructed from the panel. Each series is standardised on an expanding window that uses only months up to and including the formation month,

$$s_t = \frac{x_t - \hat{\mu}_{1:t}}{\hat{\sigma}_{1:t} + \varepsilon}, \quad (3)$$

and the standardised values are clipped to the interval $[-4, 4]$. Because the standardisation uses only past and present data, no future information enters the state, and the conditioning respects temporal integrity.

6 Evaluation and concluding remarks

On the evaluation window the model attains a Sharpe ratio of 2.373 in the volatility scaled form and 2.336 in the unscaled form. An otherwise identical model without the conditioning attains 2.484 on this panel, so the conditioning, in the disciplined form adopted here, neither improves nor materially harms the selection signal. This is consistent with the finding on this panel that the

conditional mean of returns is difficult to time, in contrast to the conditional volatility, which is timed successfully by the leverage overlay. We conclude that the state conditional model is a careful attempt to exploit the time variation of premia whose near neutral result is itself informative, since it localises the difficulty to the predictability of the conditional mean rather than to the mechanism of conditioning.